🧠 **Notes on Computational Neuroscience & EEG/BCI Concepts**

1. Foundations

- **Computational Neuroscience** → The study of brain functions using mathematical models, simulations, and computational techniques.
- **BCI (Brain–Computer Interface)** → Direct communication pathway between neural activity and external devices (prosthetics, computers).
- **Vision Perceptions** → How the brain interprets electrical signals from the retina into images.

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2. Neural Signals & Transmission

- **Electric Impulse** → Basis of neural communication.
- **Arboreal Projections** → Dendritic branches where synaptic connections form.
- **Post-Synaptic Potential (PSP)** → Excitatory or inhibitory changes in membrane potential after neurotransmitter release.
- **Action Potential** → All-or-none electrical signal along the axon.
- **All-or-None Response** → Neurons either fire fully or not at all.
- **Lossy Propagation** → Signal strength diminishes if conditions are not optimal (e.g., noise, distance).

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3. Brain Organization

- **Brain Lobes**:
  - *Frontal* → Planning, motor control
  - *Parietal* → Sensory integration
  - *Temporal* → Auditory, memory
  - *Occipital* → Vision

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4. EEG (Electroencephalography)

- **Birth of EEG** → Discovered by Hans Berger (1924).
- **Noise vs Artifacts**:
  - *Noise* → Unwanted random signals from environment/equipment.
  - *Artifacts* → Biological/movement-related disturbances (e.g., eye blink, muscle movement).
- **Sampling Rate** → Number of samples per second (Hz); critical for signal resolution.
- **Nyquist Equation** → Sampling rate ≥ 2 × maximum frequency of interest.

- **Oscillatory Brain Rhythms**:
    - Delta (0.5–4 Hz)
    - Theta (4–8 Hz)
    - Alpha (8–12 Hz)
    - Beta (12–30 Hz)
    - Gamma (>30 Hz, often 40–100 Hz, occipital for vision).
- **Good Data Collection Practices**:
    - Minimize artifacts
    - Pilot test before real data collection
    - Ensure proper electrode placement (scalp & earlobes).
- **Tools**: Net Station Acquisition, Python (MNE, SciPy), MATLAB (EEGLAB).

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5. Signal Processing & Analysis

- **ICA (Independent Component Analysis)** → Separate mixed signals (e.g., remove eye-blink artifacts).
- **PCA (Principal Component Analysis)** → Dimensionality reduction.
- **Forward Model** → Predict EEG signals from brain sources.
- **Reverse/Inverse Model** → Estimate brain sources from EEG data.
- **Feature Extraction in EEG**:
    - *Occipital Gamma* (vision tasks)
    - *Hilbert Transform* → Extract instantaneous amplitude & phase.
    - *Cumulant Maximization* → For non-linear dependencies.

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6. Neural & Network Models

- **Oscillatory Neural Networks** → Model brain rhythms & synchrony.
- **Brain vs Cardiac Signal Throughput** → Brain (~ms scale), Heart (~beat cycles).
- **Gradient Descent** → Optimization method for neural network training.
- **Rule-Based vs Perceptrons** → Symbolic AI vs early neural models.
- **Reinforcement Learning** → Trial-and-error learning (e.g., reward signals).
- **Hopfield Network** → Recurrent associative memory model.
- **Polynomial Regression** → Modeling non-linear relationships.
- **RNN (Recurrent Neural Networks)** → Sequence modeling (time-series like EEG).

- **LSTM (Long Short-Term Memory)** → Overcomes vanishing gradient, handles long dependencies.

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7. Visualization & Testing

- **Butterfly Effect Plot** → Overlay of all EEG channels to view synchronization.
- **Pilot Testing** → Small-scale trial before full-scale experiment to catch issues.
- **Scalp/Ear Lobe Coverage in Lab** → Electrode placement according to international 10–20 system.